Biological Networks Part B – Calculation Lab

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Network (Part 2): 8 proteins (A–H), 10 undirected interactions: A-B, A-C, A-D, A-E, A-H, B-C, B-G, E-F, F-G, G-H

Task 1 – Adjacency matrix

	A	B	C	D	E	F	G	H	Degree
A	0	**1**	**1**	**1**	**1**	0	0	**1**	**5**
B	**1**	0	**1**	0	0	0	**1**	0	**3**
C	**1**	**1**	0	0	0	0	0	0	**2**
D	**1**	0	0	0	0	0	0	0	**1**
E	**1**	0	0	0	0	**1**	0	0	**2**
F	0	0	0	0	**1**	0	**1**	0	**2**
G	0	**1**	0	0	0	**1**	0	**1**	**3**
H	**1**	0	0	0	0	0	**1**	0	**2**
Sum	5	3	2	1	2	2	3	2	**20**

Task 2 – Degree and hubs

Node	Degree k	Degree centrality k/(N−1)	Farness (Σ d)	Closeness 1/farness	Normalised closeness (N−1)/farness
A	5	5/7 = 0.714	9	1/9 = 0.111	7/9 = 0.778
B	3	3/7 = 0.429	11	1/11 = 0.091	7/11 = 0.636
C	2	2/7 = 0.286	13	1/13 = 0.077	7/13 = 0.538
D	1	1/7 = 0.143	15	1/15 = 0.067	7/15 = 0.467
E	2	2/7 = 0.286	12	1/12 = 0.083	7/12 = 0.583
F	2	2/7 = 0.286	14	1/14 = 0.071	7/14 = 0.500
G	3	3/7 = 0.429	12	1/12 = 0.083	7/12 = 0.583
H	2	2/7 = 0.286	12	1/12 = 0.083	7/12 = 0.583
Total	**20**		98		

Working: shortest-path distance matrix (BFS)

d	A	B	C	D	E	F	G	H	Farness
A	0	1	1	1	1	2	2	1	**9**
B	1	0	1	2	2	2	1	2	**11**
C	1	1	0	2	2	3	2	2	**13**
D	1	2	2	0	2	3	3	2	**15**
E	1	2	2	2	0	1	2	2	**12**

F	2	2	3	3	1	0	1	2	**14**
G	2	1	2	3	2	1	0	1	**12**
H	1	2	2	2	2	2	1	0	**12**

Answers to Task 2 questions

1. Sum of degrees and number of edges

$\Sigma k = 5 + 3 + 2 + 1 + 2 + 2 + 3 + 2 =$ **20**

$E = \Sigma k \div 2 = 20 \div 2 =$ **10 edges**

2. Average degree

$\langle k \rangle = \Sigma k \div N = 20 \div 8 =$ **2.500** (equivalently $2E/N = 20/8$).

3. Hub and least connected node

- **Hub: protein A** – degree 5, degree centrality 0.714. It interacts with 5 of the 7 other proteins and also has the highest closeness (0.111; normalised 0.778), so it is the best-connected node and the most likely essential protein / drug target.
- **Least connected: protein D** – degree 1 (centrality 0.143). It only interacts with A and has the largest farness (15), so it has the lowest closeness (0.067).

4. Closeness centrality

Included in the Task 2 table. Ranking (highest to lowest closeness): A (0.111) > B (0.091) > E = G = H (0.083) > C (0.077) > F (0.071) > D (0.067). A reaches every protein within 2 steps, so signals from A spread fastest; D sits at the edge of the network and is the slowest to reach the others.

Additional network properties

Property	Calculation and result
Density	$2E / [N(N-1)] = 20 / 56 = 0.357$
Average path length	$\Sigma d / [N(N-1)/2] = 49 / 28 = 1.750$
Diameter	3 (e.g. C–F, D–F, D–G)
Average clustering coefficient	(0.100 + 0.333 + 1 + 0 + 0 + 0 + 0 + 0) / 8 = 0.179
Connected components	1 – every protein can reach every other protein

Clustering coefficient per node

Node	Neighbours	k	Links among neighbours	C = links / [k(k−1)/2]
A	B, C, D, E, H	5	1	1/10 = 0.100
B	A, C, G	3	1	1/3 = 0.333
C	A, B	2	1	1/1 = 1.000
D	A	1	0	0 ($k < 2$)
E	A, F	2	0	0/1 = 0.000
F	E, G	2	0	0/1 = 0.000
G	B, F, H	3	0	0/3 = 0.000
H	A, G	2	0	0/1 = 0.000

Interpretation: about 36% of all possible interactions exist, and any two proteins are on average fewer than 2 steps apart, so signals travel efficiently. Clustering is low except in the A–B–C triangle (C has C = 1), which is the only tightly knit module. A is a hub with low clustering (0.100): it links otherwise separate branches (B–C, D, E–F, H–G), so it is also a bottleneck.